JOINS

```
CREATE TABLE DEPT(
DEPARTMENTID INT PRIMARY KEY,
DEPARTMENTNAME VARCHAR(100) NOT NULL UNIQUE,
DEPARTMENTCODE VARCHAR(50) NOT NULL UNIQUE,
LOCATION VARCHAR(50) NOT NULL)

INSERT INTO DEPT VALUES(1,'ADMIN','ADM','A-BLOCK')
INSERT INTO DEPT VALUES(2,'COMPUTER','CE','C-BLOCK')
INSERT INTO DEPT VALUES(3,'CIVIL','CI','G-BLOCK')
INSERT INTO DEPT VALUES(4,'ELECTRICAL','EE','E-BLOCK')
INSERT INTO DEPT VALUES(5,'MECHANICAL','ME','B-BLOCK')

DROP TABLE PERSON
CREATE TABLE PERSON(
PERSONID INT PRIMARY KEY,
PERSONNAME VARCHAR(100) NOT NULL,
DEPARTMENTID INT NULL FOREIGN KEY REFERENCES DEPT(DEPARTMENTID),
SALARY DECIMAL(8,2) NOT NULL,
JOININGDATE    DATETIME NOT NULL,
CITY VARCHAR(100) NOT NULL)

INSERT INTO PERSON VALUES(101,'RAHUL TRIPATHI',2,56000,'01-01-2000','RAJKOT')
INSERT INTO PERSON VALUES(102,'HARDIK PANDYA',3,18000,'25-SEP-2001','AHMEDABAD')
INSERT INTO PERSON VALUES(103,'BHAVIN KANANI',4,25000,'14-MAY-2000','BARODA')
INSERT INTO PERSON VALUES(104,'BHOOMI VAISHNAV',1,39000,'08-FEB-2005','RAJKOT')
INSERT INTO PERSON VALUES(105,'ROHIT TOPIYA',2,17000,'23-JUL-2001','JAMNAGAR')
INSERT INTO PERSON VALUES(106,'PRIYA MENPARA',NULL,9000,'18-OCT-2000','AHMEDABAD')
INSERT INTO PERSON VALUES(107,'NEHA SHARMA',2,34000,'25-DEC-2002','RAJKOT')
INSERT INTO PERSON VALUES(108,'NAYAN GOSWAMI',3,25000,'01-JUL-2001','RAJKOT')
INSERT INTO PERSON VALUES(109,'MEHUL BHUNDIYA',4,13500,'09-JAN-2005','BARODA')
INSERT INTO PERSON VALUES(110,'MOHIT MARU',5,14000,'25-MAY-2000','JAMNAGAR')

SELECT * FROM DEPT
SELECT * FROM PERSON
-------------------- SIMPLE JOINS----------------------
--1 LIST ALL PEOPLE ALONG WITH THEIR DEPARTMENT NAMES.
SELECT
        PERSON.PERSONNAME,
        DEPT.DEPARTMENTNAME
FROM PERSON
full outer JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID;



--2 SHOW EACH PERSON'S NAME, SALARY, AND DEPARTMENT.

SELECT
        PERSON.PERSONNAME,
        PERSON.SALARY,
        DEPT.DEPARTMENTNAME
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID;
```

```
--3  RETRIEVE EACH PERSON'S NAME, CITY, AND THE BLOCK LOCATION OF THEIR DEPARTMENT.

SELECT
        PERSON.PERSONNAME,
        PERSON.CITY,
        DEPT.LOCATION
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID;


--4 DISPLAY PERSONID, NAME, JOINING DATE AND DEPTNAME.
SELECT
        PERSON.PERSONID,
        PERSON.PERSONNAME,
        PERSON.JOININGDATE,
        DEPT.DEPARTMENTNAME
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID


-------------SIMPLE JOIN + WHERE--------------------


--5 DISPLAY THE NAME OF PEOPLE WORK IN THE COMPUTER DEPARTMENT

SELECT
        PERSON.PERSONNAME,
        DEPT.DEPARTMENTNAME
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE DEPT.DEPARTMENTNAME = 'COMPUTER';


--6 DISPLAY PEOPLE LIVE IN RAJKOT, AND WHAT'S THEIR DEPARTMENT?
SELECT
        PERSON.PERSONNAME,
        DEPT.DEPARTMENTNAME,
        PERSON.CITY
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.CITY = 'RAJKOT';

--7 DISPLAY PERSON NAME WHO IS SITTING IN 'C-BLOCK'
SELECT PERSON.PERSONNAME,DEPT.LOCATION
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
WHERE DEPT.LOCATION='C-BLOCK'

---------- JOIN + WHERE + MULTIPLE CONDITIONS ------------


--8 WHO EARNS MORE THAN 15,000 AND LIVES IN AHMEDABAD
```

```sql
--DISPLAY THE DETAIL WITH THEIR DEPARTMENT NAME?
SELECT
        PERSON.PERSONNAME,
        PERSON.SALARY,
        PERSON.CITY,
        dept.DepartmentName
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.SALARY > 15000 AND PERSON.CITY = 'AHMEDABAD';

--9 WHICH PERSON BELONG TO DEPARTMENTS WITH CODES 'CE' OR 'EE'?
SELECT
        PERSON.PERSONNAME,
        DEPT.DEPARTMENTCODE
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE DEPT.DEPARTMENTCODE IN ('CE','EE');

--10 WHO EARNS A SALARY NOT EQUAL TO 17,000
--AND DOES NOT LIVE IN BARODA
--DISPLAY WITH DEPTNAME.
SELECT
        PERSON.PERSONNAME,
        PERSON.SALARY,
        DEPT.DEPARTMENTNAME
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.SALARY != 17000 AND PERSON.CITY <> 'BARODA';

--11 FIND PEOPLE WITH DEPT NAME WHOSE SALARY IS BETWEEN 20,000 AND 40,000

SELECT
        PERSON.PERSONNAME,
        PERSON.SALARY
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.SALARY BETWEEN 20000 AND 40000;

--12 WHO WORKS IN ANY DEPARTMENT WHOSE NAME STARTS WITH 'C'
--AND EARNS MORE THAN 10,000?
SELECT
        PERSON.PERSONNAME,
        DEPT.DEPARTMENTNAME,
        PERSON.SALARY
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE DEPT.DEPARTMENTNAME LIKE 'C%' AND PERSON.SALARY > 10000;


--13 DISPLAY PERSON WHO JOINED THE CIVIL DEPARTMENT AFTER 1-AUG-2001

SELECT PERSON.PERSONNAME,DEPT.DEPARTMENTNAME,PERSON.JOININGDATE
```

```sql
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
WHERE DEPT.DEPARTMENTNAME='CIVIL'
AND PERSON.JOININGDATE>'1-AUG-2001'


---------JOIN + GROUP BY------------
--14 WHAT IS THE AVERAGE SALARY OF EACH DEPARTMENT

SELECT
        DEPT.DEPARTMENTNAME,
        AVG(PERSON.SALARY) AS AVGSAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME;

--15  COUNT HOW MANY PEOPLE ARE WORKING IN EACH DEPARTMENT.
SELECT
        DEPT.DEPARTMENTNAME,
        COUNT(PERSON.PERSONID) AS NUMPEOPLE
FROM DEPT
JOIN PERSON
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME;

--16 DISPLAY TOTAL SALARIES CITY WISE.
SELECT
        PERSON.CITY,
        SUM(PERSON.SALARY) AS TOTAL_SAL
FROM PERSON
GROUP BY PERSON.CITY


-- JOIN + WHERE + GROUP BY

--17 WHAT'S THE MINIMUM SALARY PER DEPARTMENT FOR PEOPLE LIVE IN RAJKOT?
SELECT
        DEPT.DEPARTMENTNAME,
        MIN(PERSON.SALARY) AS MIN_SAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.CITY = 'RAJKOT'
GROUP BY DEPT.DEPARTMENTNAME;

--18 COUNT PEOPLE IN IN EACH DEPARTMENT LIVES IN AHMEDABAD
SELECT
        DEPT.DEPARTMENTNAME,
        COUNT(*) AS NUMEMP
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.CITY = 'AHMEDABAD'
GROUP BY DEPT.DEPARTMENTNAME;
```

```sql
--19 SUM OF SALARIES FOR PERSON WHO JOINED BEFORE 2001, IN EACH DEPARTMENT.
SELECT
        SUM(PERSON.SALARY) AS TOTAL,
        DEPT.DEPARTMENTNAME
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
WHERE PERSON.JOININGDATE < '2001-01-01'
GROUP BY DEPT.DEPARTMENTNAME


-- JOIN + GROUP BY + HAVING---

--20 WHICH DEPARTMENTS HAVE AN AVERAGE SALARY HIGHER THAN 30,000
SELECT
        DEPT.DEPARTMENTNAME,
        AVG(PERSON.SALARY) AS AVGSAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME
HAVING AVG(PERSON.SALARY) > 30000;


--21 DISPLAY DEPARTMENTS WHERE MORE THAN 2 EMPLOYEES ARE WORKING.
SELECT
        DEPT.DEPARTMENTNAME,
        COUNT(*) AS NUMEMP
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME
HAVING COUNT(*) > 2;


--22 FIND ALL DEPARTMENTS WHOSE TOTAL SALARY IS EXCEEDING 100000
SELECT DEPT.DEPARTMENTNAME,SUM(PERSON.SALARY)
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME
HAVING SUM(PERSON.SALARY)>100000

--JOIN + WHERE + GROUP BY + HAVING

--23) AMONG PEOPLE IN RAJKOT,WHICH DEPARTMENTS HAVE AVG SALARY
--LESS THEN 30,000?
SELECT
        DEPT.DEPARTMENTNAME,
        AVG(PERSON.SALARY) AS AVGSAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.CITY = 'RAJKOT'
GROUP BY DEPT.DEPARTMENTNAME
HAVING AVG(PERSON.SALARY)<30000;
```

```sql
--24 IN AHMEDABAD, WHICH DEPARTMENTS HAVE ONLY ONE EMPLOYEE?
SELECT
        DEPT.DEPARTMENTNAME,
        COUNT(PERSON.PERSONID) AS NUMEMP
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.CITY = 'AHMEDABAD'
GROUP BY DEPT.DEPARTMENTNAME
HAVING COUNT(PERSON.PERSONID) = 1;

---25 GROUPING ON MULTIPLE COLUMN-----

--HOW MANY EMPLOYEES ARE THERE IN EACH CITY AND EACH DEPARTMENT?
SELECT
        PERSON.CITY,
        DEPT.DEPARTMENTNAME,
        COUNT(PERSON.PERSONID) AS NUMEMP
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY PERSON.CITY, DEPT.DEPARTMENTNAME;


--26 JOIN WITH 2 AGGREGATE FUNCTIONS
--SHOW THE PERSON COUNT AND MAXIMUM SALARY PER DEPARTMENT.
SELECT DEPT.DEPARTMENTNAME,
       COUNT(PERSON.PERSONID) AS NUMEMP,
       MAX(PERSON.SALARY) AS MAXSAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME;

--27 SHOW THE TOTAL AND AVERAGE SALARY PER DEPARTMENT.
SELECT DEPT.DEPARTMENTNAME,
       SUM(PERSON.SALARY) AS TOTALSAL,
       AVG(PERSON.SALARY) AS AVGSAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME;


--28 CONCATENATE STRING WITH JOIN
--PRODUCE OUTPUT LIKE: <PERSONNAME> LIVES IN <CITY>
--AND WORKS IN <DEPARTMENTNAME> DEPARTMENT AND EARNS <SALARY> (IN SINGLE COLUMN)
SELECT
PERSON.PERSONNAME + ' LIVES IN ' + PERSON.CITY + ' AND WORKS IN '
+ DEPT.DEPARTMENTNAME + ' DEPARTMENT AND EARNS ' +
CAST(PERSON.SALARY AS VARCHAR)
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
```

```sql
--29 UPDATE WITH JOIN
--INCREASE THE SALARY BY 10% FOR ALL EMPLOYEES WHOSE DEPARTMENT CODE 'CE'.

UPDATE PERSON
SET PERSON.SALARY = PERSON.SALARY+ (PERSON.SALARY * 0.1)

FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID

WHERE DEPT.DEPARTMENTCODE = 'CE';



---JOIN WITH ALIAS-----
--30 DISPLAY PERSON NAME AND DEPARTMENT NAME
SELECT
        P.PERSONNAME,
        D.DEPARTMENTNAME
FROM PERSON P
JOIN DEPT D
ON P.DEPARTMENTID=D.DEPARTMENTID




-------TYPES OF JOIN------
---31 INNER JOIN---
SELECT
        PERSON.PERSONNAME,
        DEPT.DEPARTMENTNAME
FROM PERSON
INNER JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID


---32 LEFT OUTER JOIN----
SELECT
        PERSON.PERSONNAME,
        DEPT.DEPARTMENTNAME
FROM PERSON
LEFT OUTER JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID

---33 RIGHT OUTER JOIN----
SELECT
        PERSON.PERSONNAME,
        DEPT.DEPARTMENTNAME
FROM PERSON
RIGHT OUTER JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID

---34 FULL OUTER JOIN----
SELECT
        PERSON.PERSONNAME,
        DEPT.DEPARTMENTNAME
FROM PERSON
```

```
FULL OUTER JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID


---35 CROSS JOIN----
SELECT *
FROM PERSON
CROSS JOIN
DEPT
--OR
SELECT *
FROM PERSON, DEPT


---SELF JOIN---
CREATE TABLE Employe(
EmpNo   VARCHAR(6)     PRIMARY KEY,
Name    VARCHAR(50)    NOT NULL,
MngrNo  VARCHAR(6)     NULL)

INSERT INTO Employe (EmpNo, Name, MngrNo) VALUES
  ('E00004', 'Milan',   NULL),
  ('E00005', 'Jay',     NULL),
  ('E00002', 'Rohan',   'E00005'),
  ('E00003', 'Priya',   'E00005'),
  ('E00001', 'Tarun',   'E00002'),
  ('E00006', 'Anjana',  'E00003');

  SELECT * FROM Employe

--36 TO RETRIEVE THE LIST OF EMPLOYEES NAME AND THEIR MANAGERS NAME
SELECT
        E.NAME AS EMPLOYE,
        M.NAME AS MANAGER
FROM EMPLOYE E
INNER JOIN EMPLOYE M
ON E.MNGRNO=M.EMPNO



-------------------3 TABLES JOIN--------------
--  AUTHORS TABLE
CREATE TABLE AUTHORS (
    AUTHORID INT PRIMARY KEY,
    AUTHORNAME VARCHAR(100),
    BIRTHDATE DATE
);
INSERT INTO AUTHORS (AUTHORID, AUTHORNAME, BIRTHDATE)
VALUES
(1, 'RABINDRANATH TAGORE', '1861-05-07'),
(2, 'CHETAN BHAGAT', '1974-04-22'),
(3, 'ARUNDHATI ROY', '1961-11-24');

-- CREATING THE PUBLISHERS TABLE
CREATE TABLE PUBLISHERS (
    PUBLISHERID INT PRIMARY KEY,
    PUBLISHERNAME VARCHAR(100),
```

```sql
    COUNTRY VARCHAR(50)
);
INSERT INTO PUBLISHERS (PUBLISHERID, PUBLISHERNAME, COUNTRY)
VALUES
(1, 'RUPA PUBLICATIONS', 'INDIA'),
(2, 'PENGUIN RANDOM HOUSE INDIA', 'INDIA'),
(3, 'HARPERCOLLINS INDIA', 'INDIA');

-- CREATING THE BOOKS TABLE
CREATE TABLE BOOKS (
    BOOKID INT PRIMARY KEY,
    TITLE VARCHAR(200),
        PUBLISHEDYEAR DATETIME,
    AUTHORID INT REFERENCES AUTHORS(AUTHORID),
    PUBLISHERID INT REFERENCES PUBLISHERS(PUBLISHERID)
);
INSERT INTO BOOKS
VALUES
(1, 'GITANJALI','1-JAN-1910', 1, 1 ),
(2, 'FIVE POINT SOMEONE','1-JAN-2004', 2, 1 ),
(3, 'THE GOD OF SMALL THINGS','1-JAN-1997',3, 2 );

--37 DISPLAY BOOK TITLE,AUTHERNAME,PUBLISHER NAME, AND PUBLISHED YEAR
SELECT
    BOOKS.TITLE,
        BOOKS.PUBLISHEDYEAR,
    AUTHORS.AUTHORNAME,
    PUBLISHERS.PUBLISHERNAME
FROM
    BOOKS
        JOIN AUTHORS
        ON BOOKS.AUTHORID = AUTHORS.AUTHORID

        JOIN PUBLISHERS
        ON BOOKS.PUBLISHERID = PUBLISHERS.PUBLISHERID;

--38 DISPLAY BOOK TITLE,AUTHOR NAME,PUBLISHED YEAR
--AND PUBLISHER'S COUNTRY
SELECT
        BOOKS.TITLE,
        AUTHORS.AUTHORNAME,
        PUBLISHERS.COUNTRY,
        BOOKS.PUBLISHEDYEAR

FROM
        BOOKS
        JOIN AUTHORS
        ON BOOKS.AUTHORID=AUTHORS.AUTHORID

        JOIN PUBLISHERS
        ON BOOKS.PUBLISHERID=PUBLISHERS.PUBLISHERID

--39 INCLUDE ALL BOOKS EVEN IF AUTHOR OR PUBLISHER IS MISSING
SELECT
  B.TITLE,
  A.AUTHORNAME,
  P.PUBLISHERNAME,
```

```
  B.PUBLISHEDYEAR
FROM BOOKS B
LEFT JOIN AUTHORS A
  ON B.AUTHORID = A.AUTHORID
LEFT JOIN PUBLISHERS P
  ON B.PUBLISHERID = P.PUBLISHERID;

--40 ALL THE DETAILS OF BOOKS PUBLISHED AFTER 1990 IN DESC ORDER

SELECT
  B.TITLE,
  A.AUTHORNAME,
  A.BIRTHDATE,
  P.PUBLISHERNAME,
  P.COUNTRY,
  B.PUBLISHEDYEAR
FROM BOOKS B
INNER JOIN AUTHORS A
  ON B.AUTHORID = A.AUTHORID
INNER JOIN PUBLISHERS P
  ON B.PUBLISHERID = P.PUBLISHERID

WHERE B.PUBLISHEDYEAR > '1990-12-31'
ORDER BY B.PUBLISHEDYEAR DESC;

----NO OF RECORDS RETURNS BY DIFFERENT TYPES OF JOINS---
CREATE TABLE T1
(NO1 INT)

INSERT INTO T1 VALUES(1),(1),(1),(2),(3),(3),(3)


SELECT * FROM T1
SELECT * FROM T2

CREATE TABLE T2
(NO1 INT)

INSERT INTO T2 VALUES(1),(1),(2),(2),(4),(NULL)




--INNER JOIN
SELECT *
FROM T1
INNER JOIN T2
ON T1.NO1=T2.NO1
select * from t1
select * from t2


--LEFT OUTER JOIN
SELECT *
FROM T1
LEFT OUTER JOIN T2
ON T1.NO1=T2.NO1
```

```
---RIGHT OUTER JOIN

SELECT *
FROM T1
RIGHT OUTER JOIN T2
ON T1.NO1=T2.NO1

---FULL OUTER JOIN--
--INNER JOIN
SELECT *
FROM T1
FULL OUTER JOIN T2
ON T1.NO1=T2.NO1

insert INTO T1 VALUES (null)
insert INTO T2 VALUES (null)
```

SET OPERATOR

```sql
CREATE TABLE ONLINE_CUSTOMER (
CID INT,
NAME VARCHAR(100)
);

INSERT INTO ONLINE_CUSTOMER VALUES(1,'MEET')
INSERT INTO ONLINE_CUSTOMER VALUES(2,'HARSH')
INSERT INTO ONLINE_CUSTOMER VALUES(3,'SHYAM')


CREATE TABLE OFFLINE_CUSTOMER (
CID INT,
NAME VARCHAR(100)
)

INSERT INTO OFFLINE_CUSTOMER VALUES(101,'MEET')
INSERT INTO OFFLINE_CUSTOMER VALUES(102,'DEV')
INSERT INTO OFFLINE_CUSTOMER VALUES(103,'SHYAM')

SELECT * FROM ONLINE_CUSTOMER
SELECT * FROM OFFLINE_CUSTOMER
-----UNION

--1 Display name of CUSTOMER who is
--either ONLINE_CUSTOMER or OFFLINE_CUSTOMER

SELECT NAME FROM ONLINE_CUSTOMER
UNION
SELECT NAME FROM OFFLINE_CUSTOMER

--- SELECT WITH ALL COLUMN
SELECT * FROM ONLINE_CUSTOMER
UNION
SELECT * FROM OFFLINE_CUSTOMER


--2 UNION ALL
--Display name of CUSTOMER who is either ONLINE_CUSTOMER
--or OFFLINE_CUSTOMER INCLUDING DUPLICATE

SELECT NAME FROM ONLINE_CUSTOMER
UNION ALL
SELECT NAME FROM OFFLINE_CUSTOMER

--3 INTERSECT
--Display name of CUSTOMER who is DOING
--bOTH ONLINE AND OFFLINE SHOPPING

SELECT NAME FROM ONLINE_CUSTOMER
INTERSECT
SELECT NAME FROM OFFLINE_CUSTOMER

--4 EXCEPT
--Display name of who IS ONLINE_CUSTOMER
--but not OFFLINE_CUSTOMER

SELECT NAME FROM ONLINE_CUSTOMER
EXCEPT
SELECT NAME FROM OFFLINE_CUSTOMER
```

```sql
--5
--Display name of  who IS OFFLINE_CUSTOMER
--but not ONLINE_CUSTOMER
SELECT NAME FROM OFFLINE_CUSTOMER
EXCEPT
SELECT NAME FROM ONLINE_CUSTOMER

CREATE TABLE EMPLOYEES(
    EMPLOYEEID INT ,
    FIRSTNAME VARCHAR(50),
    LASTNAME VARCHAR(50),
    DEPARTMENT VARCHAR(50),
    SALARY DECIMAL(10, 2)
);

INSERT INTO EMPLOYEES (EMPLOYEEID, FIRSTNAME, LASTNAME, DEPARTMENT, SALARY) VALUES
(1, 'John', 'Doe', 'HR', 60000),
(2, 'Jane', 'Smith', 'IT', 75000),
(3, 'Emily', 'Johnson', 'Finance', 80000),
(4, 'Michael', 'Brown', 'IT', 90000),
(5, 'Jessica', 'Davis', 'Marketing', 55000);

INSERT INTO EMPLOYEES VALUES(7, 'JESSICA', 'DAVIS', 'MARKETING', 55000)

SELECT * FROM EMPLOYEES
delete from EMPLOYEES where EMPLOYEEID>5
```

VIEW

```
--------VIRTUAL TABLE---------
--1 CREATE VIEW THAT SHOW FIRSTNAME,LASTNAME AND SALARY
--(COLUMN FILTER)

CREATE VIEW VW_NAME_SAL
AS
SELECT FIRSTNAME,LASTNAME,SALARY
FROM EMPLOYEES

SELECT * FROM VW_NAME_SAL

drop view VW_NAME_SAL
--2 CREATE VIEW OF IT DEPRATMENT EMPLOYEE ONLY
--(ROW FILTER)

CREATE VIEW VW_IT_EMP
AS
SELECT *
FROM EMPLOYEES
WHERE DEPARTMENT='IT'

SELECT * FROM VW_IT_EMP

drop view VW_IT_EMP
---3 CREATE VIEW THAT SHOWS FULL NAME AND SALARY OF HR DEPT

CREATE VIEW VW_FULLNAME_SALARY_HR
AS
SELECT FIRSTNAME+LASTNAME AS FULLNAME, SALARY
FROM  EMPLOYEES
WHERE DEPARTMENT='HR'

SELECT * FROM VW_FULLNAME_SALARY_HR

drop view VW_FULLNAME_SALARY_HR
--4 CREATE A VIEW THAT SHOWS THE AVERAGE SALARY OF EVERY DEPARTMENT


CREATE VIEW VW_AVG_SAL_DEPT
AS
SELECT DEPARTMENT,AVG(SALARY) AS AVG
FROM EMPLOYEES
GROUP BY DEPARTMENT

SELECT * FROM VW_AVG_SAL_DEPT

drop view VW_AVG_SAL_DEPT
--5 CREATE VIEW THAT SHOWS THE ANNUAL SALARY OF EMPLOYEE

CREATE VIEW VW_ANU_SALARY
AS
SELECT FIRSTNAME,SALARY*12 AS ANNUAL_SAL
FROM EMPLOYEES

SELECT * FROM VW_ANU_SALARY
drop view VW_ANU_SALARY
--6 CREATE VIEW OF THOSE EMP WHOSE NAME STARTS WITH 'J'
--AND SALARY IS MORE THEN 50000
```

```sql
CREATE VIEW VW_NAME_PATTERN
AS
SELECT FIRSTNAME
FROM EMPLOYEES
WHERE FIRSTNAME LIKE 'J%' AND SALARY>50000

SELECT * FROM VW_NAME_PATTERN
drop view VW_NAME_PATTERN
--7 CREATE VIEW WITH ID,FIRSTNAME, DEPARTMENT,SALARY

CREATE VIEW VW_EMPVIEW
AS
SELECT EMPLOYEEID,FIRSTNAME,DEPARTMENT,SALARY
FROM
EMPLOYEES

SELECT * FROM VW_EMPVIEW

drop view VW_EMPVIEW
-------------INSERT IN VIEW-------------
SELECT * FROM VW_EMPVIEW

--8 INSERT RECORD IN VIEW
INSERT INTO VW_EMPVIEW VALUES(10,'MILI','IT',60000)

SELECT * FROM VW_EMPVIEW

--WILL IT AFFECT THE BASE TABLE?
SELECT * FROM EMPLOYEES


-----------------UPDATE THE VIEW------------

--9 UPDATE DEPT TO 'HR' IN VIEW WHOSE ID IS 10
UPDATE  VW_EMPVIEW
SET DEPARTMENT ='HR'
WHERE EMPLOYEEID=10

SELECT * FROM VW_EMPVIEW

--WILL IT AFFECT THE BASE TABLE?
SELECT * FROM EMPLOYEES


--------------------DELETE FROM THE VIEW-------------

--10 DELETE RECORD FROM VIEW

DELETE FROM VW_EMPVIEW
WHERE FIRSTNAME='MILI'

SELECT * FROM VW_EMPVIEW

--WILL IT AFFECT THE BASE TABLE?
SELECT * FROM EMPLOYEES


----AGGREGATE FUNCTIONS,GROUP BY, HAVING CLAUSE IN VIEW----

--11 CREATE VIEW TO DISPLAY SUM OF SALARY FOR EACH DEPT

CREATE VIEW VW_SUM_SAL_DEPT
```

```
AS
SELECT DEPARTMENT,SUM(SALARY) AS TOTAL
FROM EMPLOYEES
GROUP BY DEPARTMENT

SELECT * FROM VW_SUM_SAL_DEPT

drop view VW_SUM_SAL_DEPT
--ERROR
UPDATE VW_SUM_SAL_DEPT SET
TOTAL=10000
WHERE DEPARTMENT='HR'

---WHY ERROR?
--view isn't updatable
--it's an aggregate view that uses SUM() and GROUP BY.
--In SQL, views created with aggregation or grouping are read-only



-------------CREATE VIEW WITH DISTINCT-----------------
SELECT * FROM EMPLOYEES
--12 VIEW WITH DISTINCT

CREATE VIEW VW_DISTINCTEMP
AS
SELECT DISTINCT FIRSTNAME
FROM
EMPLOYEES

SELECT * FROM VW_DISTINCTEMP

drop view VW_DISTINCTEMP
--ERROR (DUE TO DISTINCT)
--View uses DISTINCT, which makes it non-updatable.

INSERT INTO VW_DISTINCTEMP VALUES('KRITI');

--ERROR
UPDATE VW_DISTINCTEMP
SET FIRSTNAME='JOHN' WHERE
FIRSTNAME='EMILY'

-----VIEW ON BASE TABLE WITH PK------
drop table STU_INFO

CREATE TABLE STU_INFO
(ID INT PRIMARY KEY,
NAME VARCHAR(20),
SPI DECIMAL(4,2));

INSERT INTO STU_INFO VALUES(1,'MITA',5);
INSERT INTO STU_INFO VALUES(2,'RINA',7);
INSERT INTO STU_INFO VALUES(3,'KRISHNA',9);
INSERT INTO STU_INFO VALUES(4,'KRISHNA',10);


SELECT * FROM STU_INFO

---14 VIEW ON BASETABLE WITH PK
CREATE VIEW VW_NAME_SPI
AS
```

```sql
SELECT NAME,SPI FROM
STU_INFO;

SELECT * FROM VW_NAME_SPI

drop view VW_NAME_SPI
--ERROR (AS BASE TABLE CONTAINS PK)


INSERT INTO VW_NAME_SPI VALUES('MITAL',8);

-------------COMPLEX VIEW-------------
CREATE TABLE STUDENT_MASTER(
    STUDENTID INT PRIMARY KEY,
    STUDENTNAME VARCHAR(255) NOT NULL,
);

INSERT INTO STUDENT_MASTER(STUDENTID, STUDENTNAME) VALUES
(1, 'JOHN DOE'),
(2, 'JANE SMITH'),
(3, 'EMILY DAVIS');

CREATE TABLE COURSES(
    COURSEID INT PRIMARY KEY,
    COURSENAME VARCHAR(255) NOT NULL,
    STUDENTID INT FOREIGN KEY (STUDENTID) REFERENCES STUDENT_MASTER(STUDENTID)
);

INSERT INTO COURSES (COURSEID, COURSENAME, STUDENTID ) VALUES
(101,  'MATHEMATICS', 1),
(102,  'PHYSICS', 2),
(103,  'LITERATURE', 1),
(104,  'MATHEMATICS', 3);

select * from STUDENT_MASTER
select * from COURSES

---15 VIEW WITH MULTIPLE TABLE
CREATE VIEW VW_COMPLEX
AS
SELECT STUDENT_MASTER.STUDENTNAME,COURSES.COURSENAME
FROM STUDENT_MASTER JOIN
COURSES
ON STUDENT_MASTER.STUDENTID=COURSES.STUDENTID

SELECT * FROM VW_COMPLEX

drop view VW_COMPLEX
---ERROR
--is not updatable because the modification affects multiple base tables.

INSERT INTO VW_COMPLEX VALUES('MEERA','MATHS')

DELETE FROM VW_COMPLEX WHERE STUDENTNAME='JOHN DOE'

------------ALTER VIEW-------------
CREATE VIEW VW_EMP
AS
SELECT EMPLOYEEID,FIRSTNAME
FROM
EMPLOYEES
```

```sql
SELECT * FROM VW_EMP

----16 MODIFY THE EXISTING VIEW WITH ALTER
ALTER VIEW VW_EMP
AS
SELECT EMPLOYEEID,FIRSTNAME,SALARY
FROM
EMPLOYEES

drop view VW_EMP
---17 RENAME VIEW
SP_RENAME VW_EMP,VW_EMP_NEW

SELECT * FROM VW_EMP_NEW

--18 DROP VIEW
DROP VIEW VW_EMP_NEW

--19 READ ONLY VIEW--

CREATE VIEW VW_EMP_DATA
AS
SELECT EMPLOYEEID,FIRSTNAME,SALARY
FROM
EMPLOYEES

SELECT * FROM VW_EMP_DATA
SELECT * FROM EMPLOYEES

INSERT INTO VW_EMP_DATA VALUES(25,'HINA',25000);

--NOW MAKE THE VIEW READ ONLY

CREATE TRIGGER TR_VIEW
ON VW_EMP_DATA
INSTEAD OF INSERT,UPDATE,DELETE
AS
BEGIN
       PRINT 'YOU ARE NOT ALLOWED TO PERFORM DML OPERATION'
END

--NOW INSERT IN VIEW
INSERT INTO VW_EMP_DATA VALUES(30,'HEMAL',25000);

SELECT * FROM VW_EMP_DATA
SELECT * FROM EMPLOYEES

UPDATE VW_EMP_DATA SET SALARY=100
WHERE EMPLOYEEID=2

DELETE FROM VW_EMP_DATA WHERE EMPLOYEEID=2
```

SUB QUERIES

```
--NESTED QUERY OR QUERY WITH IN A QUERY--

---TYPES OF SUBQUERY---
--1 SINGLE ROW SUBQUERY (SCALAR VALUE)
--2 MUTIROW SUBQUERY
--3 CORRELATED SUBQUERY


CREATE TABLE EMP(
    EMPLOYEE_ID INT PRIMARY KEY,
    EMPLOYEE_NAME VARCHAR(100),
    SALARY DECIMAL(10, 2),
    MANAGER_ID INT,
    DEPARTMENT_ID INT,
    HIRE_DATE DATE,
    CITY VARCHAR(100)
);
INSERT INTO EMP(EMPLOYEE_ID, EMPLOYEE_NAME, SALARY, MANAGER_ID, DEPARTMENT_ID, HIRE_DATE, CITY)
VALUES
(1, 'RAJESH KUMAR', 75000.00, NULL, 1, '2010-01-15', 'MUMBAI'),
(2, 'PRIYA SHARMA', 85000.00, 1, 2, '2011-03-20', 'DELHI'),
(3, 'ANITA SINGH', 95000.00, 1, 1, '2012-05-30', 'BANGALORE'),
(4, 'VIKAS PATEL', 70000.00, 3, 3, '2013-07-10', 'CHENNAI'),
(5, 'SUNIL VERMA', 60000.00, 3, 2, '2014-09-25', 'HYDERABAD'),
(6, 'NEHA GUPTA', 90000.00, 2, 1, '2015-11-05', 'PUNE'),
(7, 'AMIT JAIN', 65000.00, 4, 3, '2016-01-20', 'KOLKATA'),
(8, 'KAVITA RAO', 80000.00, 2, 1, '2017-03-15', 'MUMBAI'),
(9, 'RAHUL DESAI', 72000.00, 1, 3, '2018-05-10', 'DELHI'),
(10, 'SUMAN MISHRA', 82000.00, 4, 2, '2019-07-30', 'BANGALORE'),
(11, 'ANITA SINGH', 95000.00, 1, 1, '2012-05-30', 'BANGALORE');


select * from emp
-------------------------------------------------------
              --SINGLE ROW SUBQUERY
-------------------------------------------------------
--1 RETRIEVE THE NAME OF THE EMPLOYEE WITH THE HIGHEST SALARY

--ERROR
SELECT EMPLOYEE_NAME,MAX(SALARY) FROM EMP

SELECT MAX(SALARY) FROM EMP
--95000

SELECT EMPLOYEE_NAME, SALARY
FROM EMP
WHERE SALARY =95000

SELECT EMPLOYEE_NAME, SALARY
FROM EMP
WHERE SALARY = (SELECT MAX(SALARY) FROM EMP);

--2 GET THE NAME OF THE EMPLOYEE WITH THE LOWEST SALARY
SELECT EMPLOYEE_NAME ,SALARY
FROM EMP
WHERE SALARY = (SELECT MIN(SALARY) FROM EMP);



--3 FIND THE EMPLOYEE WITH THE EARLIEST HIRE DATE
```

```
--RETRIEVE THE NAME OF THE EMPLOYEE WHO HAS BEEN
--WORKING THE LONGEST (BY HIRE_DATE)

--MIN --> OLDEST DATE
--MAX--> LATEST OR MOST RECENT DATE



SELECT EMPLOYEE_NAME,HIRE_DATE
FROM EMP
WHERE HIRE_DATE = (SELECT MIN(HIRE_DATE) FROM EMP);



--4 FIND THE EMPLOYEE WHO IS RECENTLY HIRED
SELECT EMPLOYEE_NAME,HIRE_DATE
FROM EMP
WHERE HIRE_DATE = (SELECT MAX(HIRE_DATE) FROM EMP);

--5 GET EMPLOYEES WHO EARN MORE THAN THE AVERAGE SALARY;


SELECT EMPLOYEE_NAME, SALARY
FROM EMP
WHERE SALARY > (SELECT AVG(SALARY) FROM EMP);

select * from emp
--6 FIND EMPLOYEES IN THE SAME DEPARTMENT AS 'NEHA GUPTA'
SELECT *
FROM EMP
WHERE DEPARTMENT_ID =
(SELECT DEPARTMENT_ID FROM EMP WHERE EMPLOYEE_NAME='NEHA GUPTA')

-----------FREQUENTLY ASKED INTERVIEW QUERY-------------

--7 FIND SECOND HIGHEST SALARY
select salary from emp
---METHOD:1
--STEP 1: ALL THE SALARY OTHER THEN MAX SALARY
SELECT SALARY
FROM EMP
WHERE SALARY<(SELECT MAX(SALARY) FROM EMP)
                    --95000

--STEP 2 FIND MAX SALARY FROM OTHER THEN MAX SALARY
SELECT MAX(SALARY)
FROM EMP
WHERE SALARY<(SELECT MAX(SALARY) FROM EMP)


---METHOD:2
--WITH ORDER BY
SELECT SALARY
FROM EMP
WHERE SALARY<(SELECT MAX(SALARY) FROM EMP)
ORDER BY SALARY DESC

---THEN SELECT TOP 1
SELECT TOP 1 SALARY
FROM EMP
```

```
WHERE SALARY<(SELECT MAX(SALARY) FROM EMP)
ORDER BY SALARY DESC


--8 SELECT N TH HIGHEST SALARY
--LET'S CONSIDER 4TH HIGHEST SALARY

--STEP 1
SELECT SALARY
FROM EMP
ORDER BY SALARY DESC

--STEP 2
SELECT DISTINCT TOP 4 SALARY
FROM EMP
ORDER BY SALARY DESC

--STEP 3
SELECT TOP 1 SALARY FROM
              (SELECT DISTINCT TOP 4 SALARY
                     FROM EMP
                     ORDER BY SALARY DESC) AS RESULT
ORDER BY SALARY

---OR----
SELECT MIN(SALARY) FROM
              (SELECT DISTINCT TOP 4 SALARY
              FROM EMP
              ORDER BY SALARY DESC) AS RESULT




SELECT EMPLOYEE_NAME,SALARY,CITY FROM EMP WHERE
SALARY=
(SELECT MAX(SALARY) FROM EMP)

--9 FIND THE EMPLOYEE WORKING IN THE SAME CITY
--AS THE HIGHEST-PAID EMPLOYEE


SELECT EMPLOYEE_NAME, CITY, SALARY
FROM EMP
WHERE CITY = (
    SELECT DISTINCT CITY
    FROM EMP
    WHERE SALARY = (SELECT MAX(SALARY) FROM EMP)
);                                --95000


select * from emp

SELECT EMPLOYEE_NAME, CITY, SALARY
FROM EMP
WHERE CITY = (
    SELECT top 1 CITY
    FROM EMP
    order by salary desc)
-------------------------------------------------------
              --MULTI ROW SUBQUERY
-------------------------------------------------------
--GET EMPLOYEES WHO WORK IN DEPARTMENTS WHERE THE AVERAGE SALARY
```

```
--OF DEPT IS ABOVE 70,000

SELECT DEPARTMENT_ID,AVG(SALARY)
FROM EMP
GROUP BY DEPARTMENT_ID
HAVING AVG(SALARY)>70000



SELECT EMPLOYEE_NAME,SALARY,DEPARTMENT_ID
FROM EMP
WHERE DEPARTMENT_ID IN
                                (SELECT DEPARTMENT_ID
                                FROM EMP
                                GROUP BY DEPARTMENT_ID
                                HAVING AVG(SALARY)>70000)

                                --1 2



--GET EMPLOYEE NAME WHO WHORKS IN A DEPARTMENT WHERE
--TOTAL EMPLOYEE IN DEPT IS MORE THAN 2

--STEP 1
SELECT DEPARTMENT_ID,count(*)
FROM EMP
GROUP BY DEPARTMENT_ID
HAVING COUNT(EMPLOYEE_NAME)>2

--STEP 2
SELECT EMPLOYEE_NAME
FROM EMP
WHERE DEPARTMENT_ID IN (SELECT DEPARTMENT_ID
                                    FROM EMP
                                    GROUP BY DEPARTMENT_ID
                                    HAVING COUNT(EMPLOYEE_NAME)>2)
                                    --1 2 3
-------------------------------------------------------
              --CORELATED ROW SUBQUERY
-------------------------------------------------------
CREATE TABLE EMPLOYEEMASTER
(EID INT,
NAME VARCHAR(10))

INSERT INTO EMPLOYEEMASTER VALUES
(1,'JAY'),(2,'MEET'),(3,'MINAL'),(4,'MIRA'),(5,'RAHUL')


CREATE TABLE DEPTMASTER
(DEPT_ID VARCHAR(5),
DEPT_NAME VARCHAR(10),
EID INT)


INSERT INTO DEPTMASTER VALUES
('D1','IT',1),('D2','HR',2),('D3','MARKETING',3)

SELECT * FROM EMPLOYEEMASTER
SELECT * FROM DEPTMASTER
```

```
--THE EXISTS OPERATOR IS USED TO TEST FOR THE EXISTENCE OF ANY RECORD IN A SUBQUERY.

--THE EXISTS OPERATOR RETURNS TRUE IF THE SUBQUERY RETURNS ONE OR MORE RECORDS.

SELECT * FROM EMPLOYEEMASTER
WHERE EXISTS
             (SELECT EID FROM DEPTMASTER
             WHERE
             EMPLOYEEMASTER.EID=DEPTMASTER.EID)
```